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Blue Mirror

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Correspondence:
Building a European Water Culture: the EU Water Academy
p. 3

Monthly Bulletin
p. 16

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Cover Image
"Inhabit the sea", pencil drawing.
Blue Mirror Editorial Team.

EDITORIAL
1 Macrotheme: *"Dwelling"*

CORRESPONDENCE
3 Building a European Water Culture: the *EU Water Academy*

COMMENT
7 COP30 in the Amazon: Science, Climate Justice and the Politics of a Warming Planet

GLOWACON NEWS
10 Strengthening Regional Capacity for Wastewater and Environmental Surveillance: Insights from the CBE Meeting in Istanbul

THE HUMAN SAMPLE
12 Previewing the XIX World Water Congress: Innovation and Adaptation in Marrakech

BRIEF COMMUNICATIONS
13 News

EU WASTEWATER OBSERVATORY FOR PUBLIC HEALTH
16 **MONTHLY BULLETIN**

EDITORIAL
Macrotheme: "Dwelling"

Bernd Manfred Gawlik

The verb *to dwell* comes from the Latin *habere*. Yet its meaning goes far beyond simply owning a home. In Latin, *habere* also means to remain, to linger in a place; and when you linger, you *gain time*. Time to learn, to understand, to grasp the subtleties. Figuratively, dwelling becomes a form of cultivation of thoughts, projects, ideas. To dwell is to grant oneself duration. To stay long enough to perceive nuances, to explore depths, to avoid consuming immediately what we encounter, and instead to understand it, to enter into it. In the end, we dwell only in what endures, in what withstands the passing of time, or in what forces us to slow it down. Kafka once wrote in a brief story that no one would think of gathering an audience merely to *crack nuts*, but if someone did, and succeeded, it might reveal that even this simple act could be an art we had long taken for granted. For me, this art is *dialogue*: a practice that requires *inhabiting* a space together with another. To dwell, here at the JRC and within *Blue Mirror*, means to engage in dialogue through method: the method of learning. Learning is not about discovering what lies beyond what we know; it is about building upon continuity, identifying cracks in our understanding, filling them, and constructing further knowledge on a scientific foundation, with a touch of vision. This is what Einstein did: he took the physics of Galileo and Maxwell and sought a framework in which everything aligned, and in doing so, discovered that "time" itself was not as we had imagined. This is the continuity of scientific history we are called to preserve. It is the purpose of *Blue Mirror* and of flagship JRC initiatives such as the *EU Water Academy*, to which JRC Director D Alessandra Zampieri contributes a special reflection in this issue, presenting the inception meeting of December 8th.

We will never fully exhaust the totality of knowledge. Yet we embrace the limits of our understanding, and upon this continuous updating and study, we base our current activities. Dwelling or lingering, acknowledges that the *Lumières* idea of a pure, disembodied reason capable of illuminating and comprehending all, no longer belongs to our era. We are within the world, not outside it; yet we strive to pass through that *small doorway* depicted on our cover: a doorway set within a beautiful seashell, opening us to future water challenges, guided by knowledge of the past and present achievements.

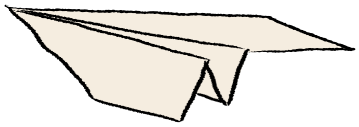
As Bachelard writes in *The Poetics of Space*: "**The house of the future may be more solid, more luminous, more spacious than all the houses of the past**." In these complex times, marked by wars and crises of every kind, we must keep our attention on this house of the future: a European home that is inclusive, curious, and willing to dwell in dialogue.

EU Water Academy

LOADING...

CORRESPONDENCE

Building a European Water Culture: the EU Water Academy



Alessandra Zampieri, Director - JRC.D, Sustainable Resources, Ispra, Italy

In the intensity of our daily work, it is easy to lose ourselves. As the writer David Foster Wallace once observed, there are moments when we feel *we have gained, and then lost, something infinite*. Periods of intense commitment, filled with prestigious projects, high-level meetings, and demanding expectations, can leave us both enriched and momentarily emptied. Yet this sensation carries a paradoxical vitality: it brings us back to the starting line. And as Isaac of Nineveh teaches in his *Ascetic Writings*, the most meaningful stance we can take in the world is that of the learner. Still, this return to the beginning naturally brings questions: am I focusing my time on the right things? Am I where I intended to be? These questions arise most forcefully in times of transition, and they often illuminate the threshold of genuine change. Twenty years ago, addressing students at Kenyon College, David Foster Wallace offered a striking insight: **education is not the ability to think, but the ability to choose what to think about**. At the JRC we have chosen to think about legacy, about impact, about the responsibility to shape a future in which knowledge matters; and for us, today, this reflection converges on something I find crucial and necessary: the institution of the *EU Water Academy*. On 8 December, we will officially launch its inception process, a crucial milestone for Europe's integrated, science-based approach to water.

This comes in a period dense with global events such as COP30 negotiations, the Mediterranean Pact, the *Ecomondo* event, the success of the LIFE Programme for environment and climate action. Amid these, we must keep our attention firmly on water, one of the most strategic, vulnerable, and foundational resources for Europe's resilience.

Why an EU Water Academy, and why now?

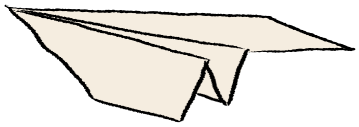
As Europe prepares for the possibility of *Diseases X* and other emerging pathogens, water systems become frontline infrastructures: (i) reservoirs of microbial evolution, (ii) pathways for transmission, (iii) essential components of surveillance and resilience. In addition, climate projections show that water availability will change fundamentally in the coming decades. River regimes will shift, with more water in winter and less in summer, under climatic conditions unlike any we have previously experienced. The *EU Water Academy* arises precisely to meet this moment. Its mission is:

I. to train experts capable of responding to tomorrow's water challenges;

II. to offer hand-on training in monitoring pathogens through wastewater via *The Wastewater Surveillance Academy for Health*

CORRESPONDENCE

Building a European Water Culture: the EU Water Academy



Alessandra Zampieri, Director - JRC.D, Sustainable Resources, Ispra, Italy

III. to foster a culture of critical, multidisciplinary, and forward-looking water reuse governance;

IV. to strengthen Europe's international dialogue on water security.

As the President of the European Commission, Ursula von der Leyen, recently put it, "We are living in turbulent times, and when economic uncertainty meets geopolitical volatility, partners like us must come closer together". This uncertainty demands places of exchange, training, and global conversation. The EU Water Academy aims to be one of those places, bringing together European and international institutions, researchers, practitioners, and policymakers.

A global community, shaped by difference, united by science

The beauty of global scientific communities is not uniformity, but difference. Discussing water with a colleague from India is not the same as discussing it with one from Germany, and it should not be. Each brings a distinct cultural lens, a different scent, a different memory of water. The Academy will give space to these differences. Yet at the same time, the Academy will stand firmly on what unites us: **the universal method of science**. Physicist Carlo Rovelli reminds us that science advances not through opinion, but through continuous refinement of past understanding. Copernicus could redraw the Earth as revolving around the Sun, only because he stood on Ptolemy's shoulders. **Innovation emerges from the careful observation of where things no longer align,** and the courage to improve them. This is the culture we want to cultivate: a European culture of water knowledge, built on continuity, curiosity, and collective intelligence.
Building a European water culture.
On 8 December, we will open a new chapter. A chapter in which Europe chooses what to think about, and decides to think deeply, scientifically, and collaboratively about water: our shared element, our shared responsibility, and our shared future.



EU Water Academy

Inception meeting
EU Water Resilience Forum
Brussels, Belgium

COMMENT

COP30 in the Amazon: Science, Climate Justice and the Politics of a Warming Planet

Giulia Panzarella

COP30 opened on 10 November in Belém, Brazil, the gateway to the Amazon, and ran until 21 November. After three consecutive COPs in major oil-producing countries such as Egypt, the United Arab Emirates, and Azerbaijan, the climate negotiations moved to the edge of the world's largest tropical rainforest. Journalist Emanuele Cotugno described this decision with a "Fitzcarraldo-like" metaphor: an audacious, improbable operation in the heart of the Amazon, echoing Werner Herzog's film about trying to stage an opera house in the Brazilian forest. The intention was clear: to confront global leaders with the heart of the climate crisis, the environmental and social conditions of northern Brazil, where deforestation, extractive industries, and indigenous resistance intersect. Tropical forests have long been debated in COP negotiations; organising one within the forest forces negotiators to witness the reality firsthand. Belém's COP has been, in many ways, the return of democracy to the negotiations, democracy with its tensions, contradictions, and uncomfortable truths. Brazil has invested heavily in oil and gas exploration, including areas near the Amazon. Indigenous communities have been protesting these developments, making their voices heard inside and outside the COP venue.

The Central Issue: Climate Finance

Every COP revolves around a single, persistent dilemma: "How can we mobilise the financial resources needed to confront the climate crisis?". Azerbaijan's COP29 ended with a compromise that many considered a failure. The proposed sum of USD 300 billion a year from 2035 for countries of the Global South was largely symbolic and far from what science indicates is needed. To align with the real scale of the climate challenge, annual finance must reach USD 1.3 trillion. This is the target of the institutional "Baku to Belém Roadmap", and one of COP30's main objectives: **elevate climate finance to scientific reality, not political convenience**; **put adaptation back at the centre**; **agree on a global definition of adaptation**; **negotiate the phase-out of fossil fuels**; **and commit to a genuine mitigation pathway**. The tone of the COP was partly set by Bill Gates, who urged negotiators to prioritise adaptation, citing health, agriculture, and resource security. The 2025 Lancet Countdown Report provides a sobering perspective: one person dies every minute due to climate-related impacts.

COMMENT

COP30 in the Amazon: Science, Climate Justice and the Politics of a Warming Planet

Giulia Panzarella

Figure 1. Demonstrators in Belém during COP30.



Source*: Il Domani*

COP of the Absences and COP of the Truth

Major emitters such as the United States, China, and India are largely absent from COP30. In the Brazilian press, this COP has been called “La COP da Verdade”, the COP of truth, highlighting both the scientific urgency and the global political divide. Despite past setbacks, hope remains.
A key intervention came from Bill Hare, physicist and founder of Climate Analytics, who has analysed emissions for decades. He described a plausible, science-based scenario: with robust mitigation and rapid fossil fuel phase-out, it is still possible to limit warming to below 1.5°C by the end of the century. Scientific realism matters. So does scientific optimism. *Defeatism is as damaging as denialism*. There is still a margin for action, but it is rapidly narrowing.

Brazil’s Climate Contradiction

President Lula da Silva arrived at COP30 with the ambition of restoring Brazil’s reputation as a climate leader. Yet Brazil embodies the contradictions of the global crisis: (i) it has joined the OPEC+ alliance and discovered the largest offshore oil field in 25 years; (ii) Petrobras, the national oil company, is expanding aggressively; (iii) Brazil now produces more oil than Azerbaijan. At the same time, renewable energy capacity is growing rapidly, but fossil fuel production is not declining it is added to the system rather than replaced. This mirrors a global pattern: we are adding clean energy without removing fossil fuels. Climate goals require substitution.

COP30 as the “People’s COP”

The Amazon is home to hundreds of Indigenous communities. Belém has infused COP30 with civic and cultural vibrancy. A flotilla of Indigenous boats travelled 3,000 km along Amazon rivers to bring demands directly to negotiators. The visible conflict between Indigenous voices and political power has been constructive, forcing negotiators to confront the human

COMMENT

COP30 in the Amazon: Science, Climate Justice and the Politics of a Warming Planet

Giulia Panzarella

dimension of the climate crisis. Indigenous leaders emphasised: "We are 5% of the world's population but we protect 80% of global biodiversity. "Indigenous territories are climate solutions; direct financing mechanisms are urgently needed; protecting forests means protecting peoples, not only trees. **Célia Xakriabá**, the first Indigenous woman elected to Brazil's federal parliament, articulated this powerfully: "The forest is not just trees. The forest is us." The economic argument is equally compelling. A study by the University of the Amazon shows that Brazil's agribusiness sector benefits by 57% from Indigenous protection of ecosystems. If destructive legislation continues, the projected economic deficit could reach USD 338 billion.

Figure 2. Célia Xakriabá



Source: *Wikipedia*

Emissions: The Scientific Reality

The UNFCCC's 2025 report suggested that global emissions might be starting to curve downward for the first time since 1990. But a recent article in *Nature* warns that emissions are still rising, albeit more slowly. Climate scientist Giulio Betti (CNR) explains that: "emissions continue to increase; what we observe is a slowdown in growth, not an actual decline; the first meaningful global reduction is only expected around 2030"; late, but crucial. Emission calculations rely on both national reporting and indirect estimates derived from energy production. The margin of uncertainty highlights the need for independent and transparent monitoring systems.

Climate, Indigenous Rights, and Legal Innovations

Indigenous demands at COP30 include:

I. official recognition of Indigenous territories,
II. direct climate financing,
III. new mechanisms such as a *Biome Fund for Brazil's six biomes* (now including coastal-marine systems).

This COP also hosted an unprecedented legislative action: female parliamentarians from multiple countries simultaneously

COMMENT

COP30 in the Amazon: Science, Climate Justice and the Politics of a Warming Planet

Giulia Panzarella

deposited a bill to map environmental and gender-based violence associated with mega-projects ("Without women there is no climate").
Meanwhile, Natalia Greene, *Director of the Global Alliance for the Rights of Nature*, recalled how "Ecuador became the first country to recognise legal rights for nature", a concept whose time, she argued with Victor Hugo, "has come."

Governing the Planet With Science and Justice

COP30 reveals the political, scientific, and moral contradictions of our time.
It takes place where the climate crisis is both visible and personal.
It shows that:
- we must accelerate finance,
- commit to deep mitigation,
- centre adaptation,
- protect defenders of ecosystems,
- uphold climate justice.

Above all, COP30 reminds us that climate governance is ultimately a science of choices: choices about energy, ecosystems, equity, and the future of the planet.
The Amazon, with its tensions and its voices, forces us to confront those choices honestly.



GLOWACON NEWS
Strengthening Regional Capacity for Wastewate
Environmental Surveillance:
Insights from the CBE Meeting in Istanbul

Bernd Manfred Gawlik

On 18-19 November 2025, Istanbul hosted a pivotal regional meeting focused on enhancing wastewater and environmental surveillance (WES) capacities across Islamic Development Bank (IsDB) member countries. This meeting brought together public health, environmental, and policy leaders to accelerate the implementation and scale-up of WES programs leveraging global expertise and resources. The meeting aimed to boost technical expertise, provided practical guidance on WES program design, including site selection, pathogen prioritization, and data integration, fostering a vibrant regional network linking health and environmental sectors. A significant goal was to create a policy roadmap to integrate WES effectively into national disease surveillance strategies. The opening session featured thought leaders from the Turkish Water Institute, IsDB, European Commission, Gates Foundation, and other organizations, highlighting the global commitment to wastewater surveillance as an innovative tool to protect health. Country showcases from Turkey, Germany, South Africa, and Indonesia presented diverse perspectives and experiences in WES program development, reflecting varied regional contexts and challenges.

Panel discussions examined global and regional opportunities for capacity strengthening, as well as critical issues around financing WES programs. Representatives from IsDB, the World Bank, Asian Development Bank, and others shared insights on how multilateral funding can scale impactful WES initiatives. An interactive World Café format allowed small groups to rotate through themed discussions on topics like strategic national WES planning, translating data into actionable insights, technical support and tailored capacity building, and securing sustainable financing. This format fostered cross-country knowledge sharing and helped map critical gaps and opportunities to inform future investment. The meeting closed with a pathogen prioritization simulation exercise moderated by experts from the Institute for Health Metrics and Evaluation and the World Health Organization, contextualising the disease burden framework essential for targeted WES surveillance.

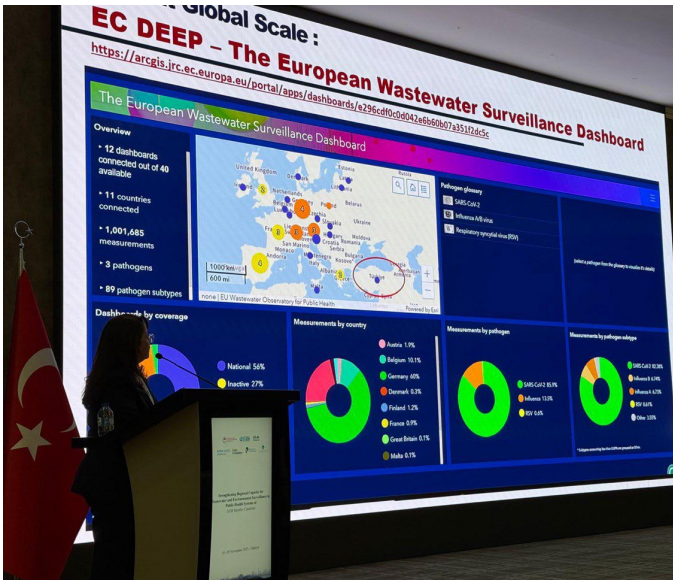
Wastewater and environmental surveillance are set to become cornerstone tools in early detection and response for emerging health threats, this meeting represented a key moment in that journey.

GLOWACON NEWS
Strengthening Regional Capacity for Wastewate Environmental Surveillance: Insights from the CBE Meeting in Istanbul

Bernd Manfred Gawlik

Figure 3. A four-image portrait featuring photographs from the conference (A and B) and from the wastewater treatment plant (C and D).









Source: *EC-JRC*

THE HUMAN SAMPLE
Previewing the XIX World Water Congress: Innovation and Adaptation in Marrakech

Bernd Manfred Gawlik

The XIX World Water Congress will take place in Marrakech, Morocco, from 1 to 5 December 2025, bringing together experts, institutions, and stakeholders from around the globe. This edition will focus on the theme “Water in a Changing World: Innovation and Adaptation,” highlighting the urgent need for new solutions and resilient strategies in the face of global change.

This special issue of Blue Mirror offers an exclusive preview of the video that will be presented during the conference, providing a first look at the reflections and innovations that will shape the discussions in Marrakech.

Video 1. *One World, One Water* Video



Source: *EC-JRC*

BRIEF COMMUNICATIONS

THE EUROPEAN WASTEWATER SURVEILLANCE DASHBOARD

The EU Wastewater Observatory for Public Health has launched a new tool: the European Wastewater Surveillance Dashboard. This innovative platform tracks the spread of infectious diseases by analysing wastewater across Europe, providing real-time insights on pathogen trends and emerging health threats to support rapid responses to disease outbreaks.



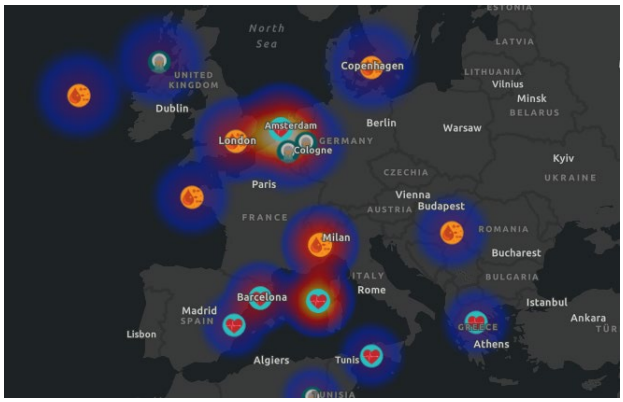
Try the tool and help us improve it - we invite you to explore the dashboard and send your feedback

ENCYCLOPAEDIA CLOACAE

Stay up-to-date with the latest information on pathogens and public health indicators in wastewater through the EU Wastewater Observatory's Encyclopaedia Cloacae.
This valuable resource supports disease detection and public health efforts across Europe.

Get involved and contribute to the growth of the Encyclopaedia by submitting publications or your organisation's logo to become an official contributor.
Our team will review and add your contributions to the platform. To contribute, please visit the "Add Your Knowledge" area.

ADVANCED HEATMAP DASHBOARD FOR WASTEWATER PATHOGEN RESEARCH MONITORING



Explore global research on pathogens in human excreta and wastewater through an interactive dashboard, featuring a dynamic heatmap of active research areas.

BRIEF COMMUNICATIONS



Are you ready to contribute to Europe's health preparedness? The European Commission is inviting expressions of interest from locations committed to using wastewater surveillance as a tool for early detection of health threats. If your site can support regular sampling—especially near transport hubs—and aligns with national or local surveillance efforts, you could become part of the European Super-Sites network. The Commission covers all sample dispatch and analysis costs and supports partnerships with key sectors.

Submit your **interest** now via the **Super-Site call** for expression of interest:
👉https://ec.europa.eu/eusurvey/runner/EU4S_SuperSites

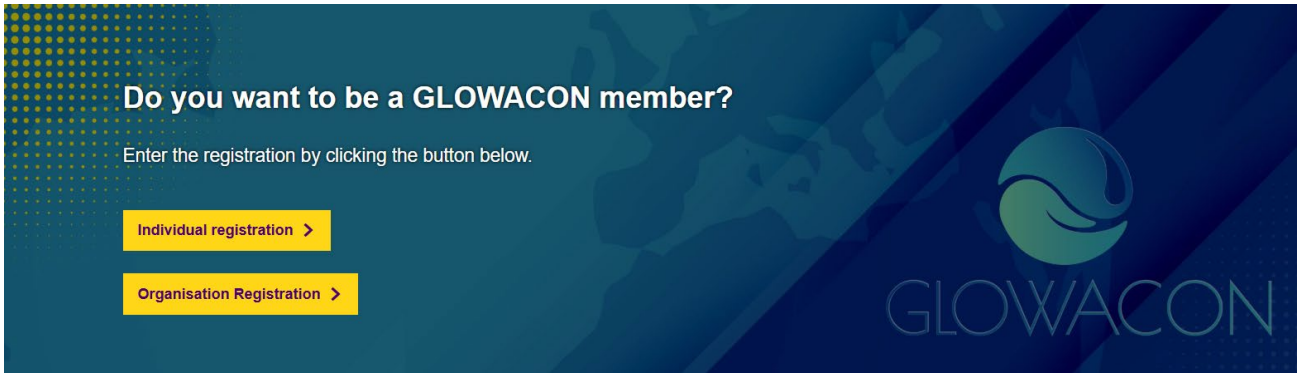
GLOWACON

GLOWACON Aviation Wastewater Surveillance Programme



With the increasing interconnectivity of global travel, the aviation sector presents a significant focus for innovative public health surveillance methods. This initiative assembles public health services, airlines, international organizations, airport authorities, academic institutions, and contracted service providers to enhance pathogen monitoring through wastewater analysis.
Through the GLOWACON Aviation Programme, we aim to address the current limitations in public health surveillance by working with the aviation sector, demonstrating the practicality and effectiveness of wastewater-based epidemiology. The insights gained will advance WES/WBE as a standard component of international public health efforts, ensuring rapid pathogen detection in a globally mobile population. Participants will not only contribute to global health safety but also benefit from cutting-edge research and data analysis opportunities, while staying in control of the derived data. Join the future of Wastewater Surveillance.

BRIEF COMMUNICATIONS

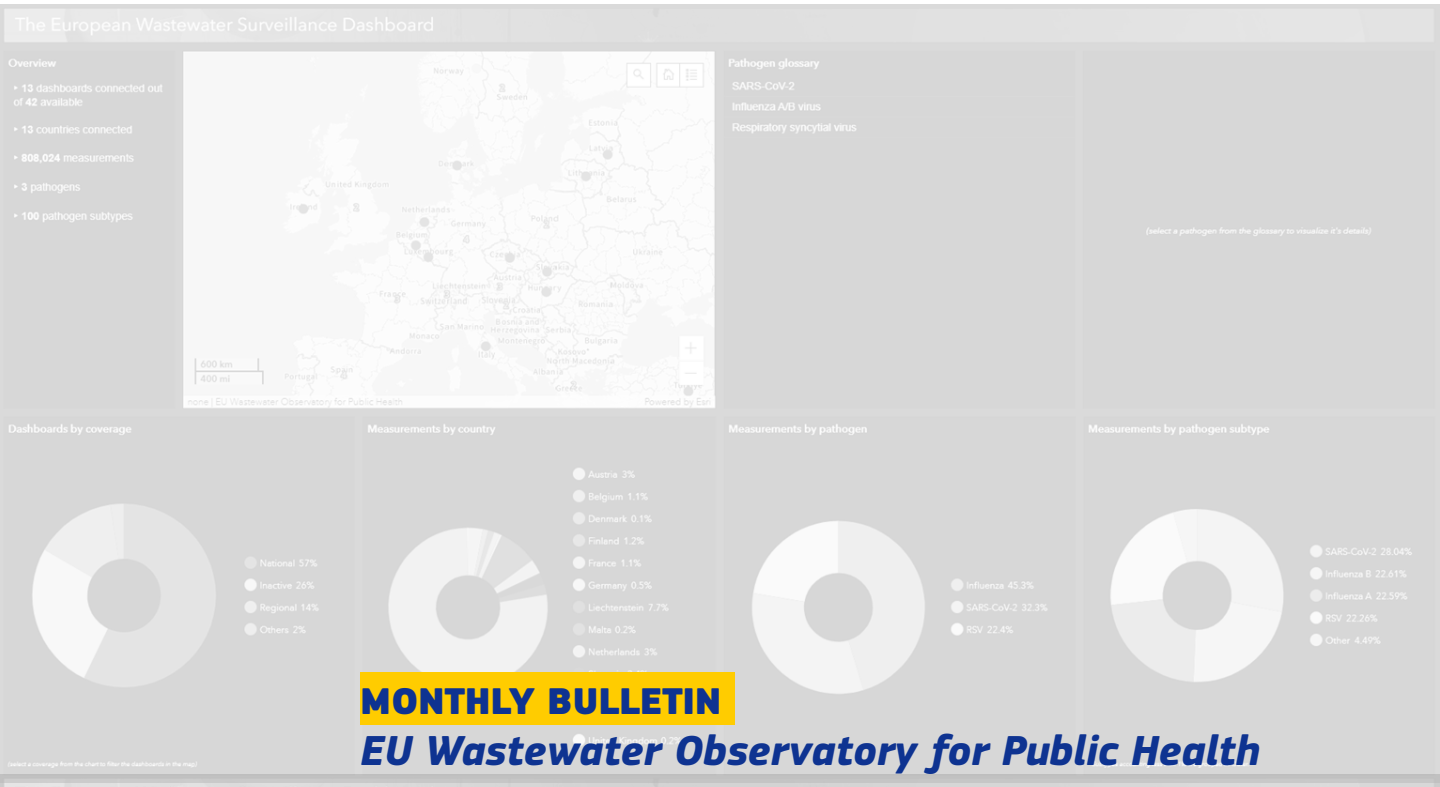


Individual registration
Organisation Registration



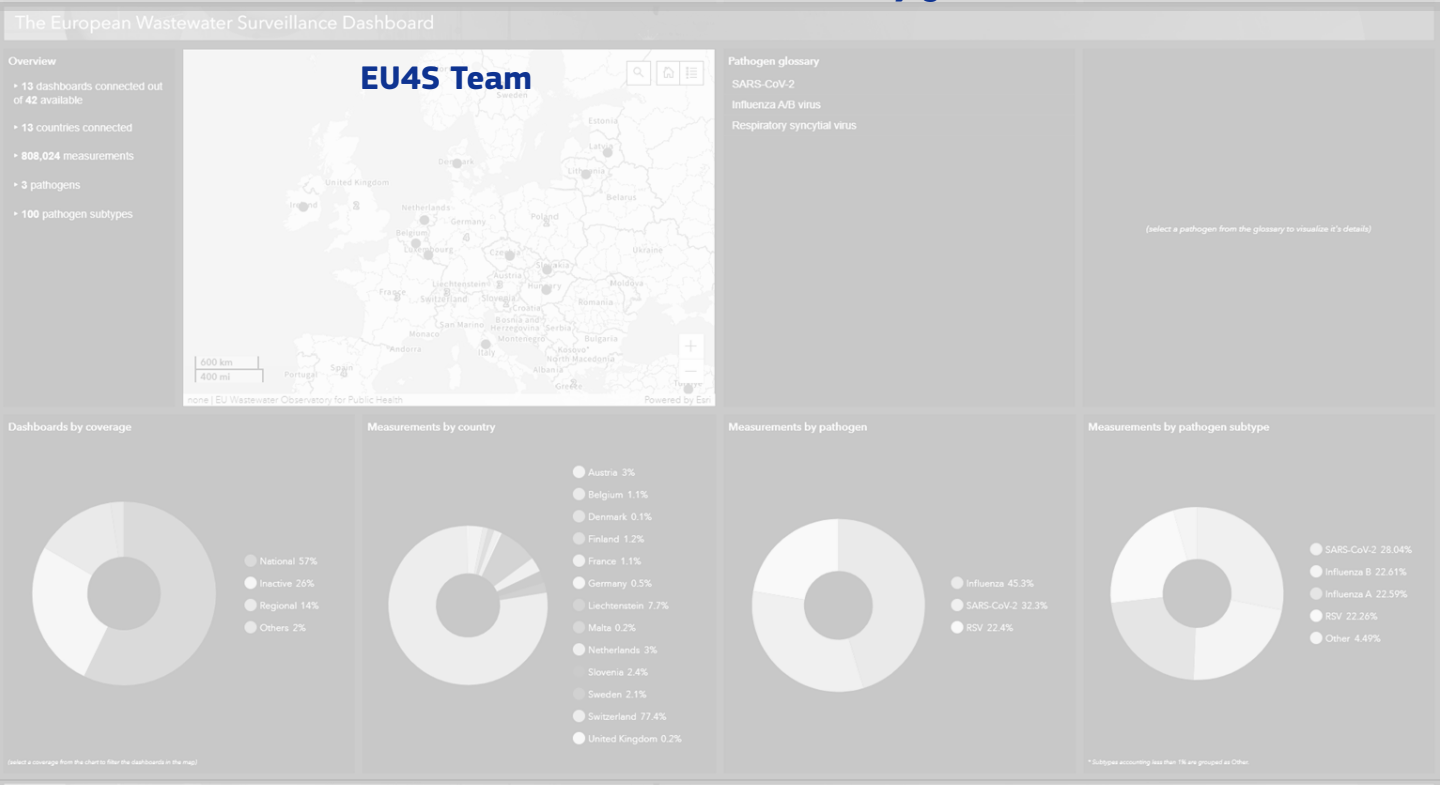
A flagship initiative to strengthen water resilience in Europe The European Union is grappling with substantial water challenges due to climate change, unsustainable practices, ageing infrastructure, and mismanagement.
In response, the European Commission has introduced the European Water Resilience Strategy, with the water efficiency first principle, featuring EUWA, the European Water Academy, as one of its flagship initiatives.
Stay tuned for further announcements.

Check out the EUWA Fact Sheet.

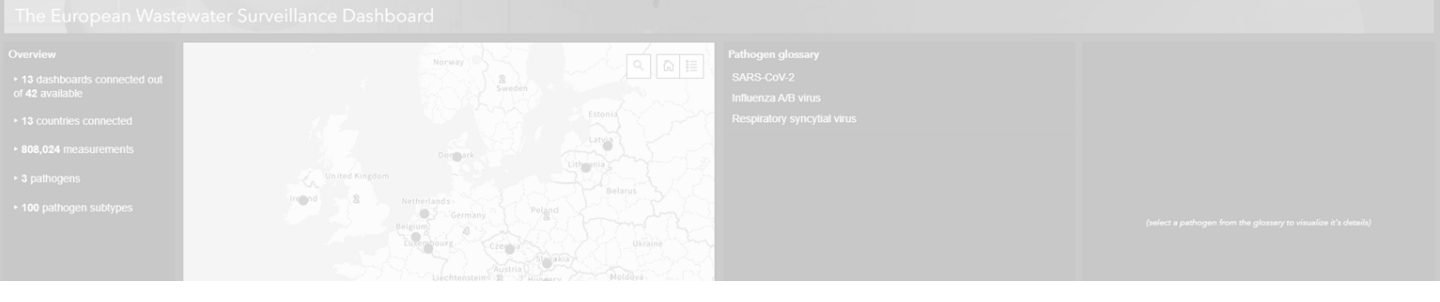


MONTHLY BULLETIN

EU Wastewater Observatory for Public Health



EU4S Team



MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

EU4S Team

EXECUTIVE SUMMARY

- **Updated SARS-CoV-2 wastewater data are available at the national level for 13 of the 27 EU countries.**
- **As of November 2025, two countries (Austria, Finland) have reported increasing trends in SARS-CoV-2 levels in wastewater. Germany and Luxembourg have reported stable levels, while Belgium, Denmark, Greece, Latvia and the Netherlands have reported decreasing levels. Viral loads are classified as moderate in Luxembourg, Austria, and Denmark; low in Belgium.**
- **In September 2025, the XFG variant (Stratus) became the most dominant lineage of SARS-CoV-2 in Europe, quickly surpassing other Omicron subvariants. At this time, there is no evidence to suggest that XFG subvariants are more likely to cause severe illness compared to previous Omicron subvariants.**
- **Influenza and RSV continue to be monitored in Austria, Belgium, Finland, Germany, Hungary, Luxembourg and Sweden. Belgium, Finland, Ireland and Slovenia are also conducting tests for poliovirus but currently no new data have been reported. Sweden and Luxembourg conducted surveillance for enteric viruses until mid-2024.**
- **Dengue, chikungunya, Zika, and yellow fever are emerging as significant arboviral diseases of growing concern. However, at present, only sporadic wastewater surveillance data are available.**
- **The EU Wastewater Observatory for Public Health is keenly following these developments and will integrate new insights in forthcoming bulletins, with input from the EU Super-Sites.**

This bulletin offers insights derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities. Additionally, data from existing regional/local, and non-EU sources are included. The most recent retrieval was performed on *30th November 2025*.

Wastewater surveillance analyses samples to reveal community health and environmental trends, enhancing traditional public health monitoring; the EU Wastewater Observatory collects and shares this data to support decision-making and pandemic preparedness, tracking viruses like SARS-CoV-2, avian flu, mpox, RSV, and influenza. In June 2025, the European Commission's Health Emergency Preparedness and Response Authority (HERA), in partnership with the Joint Research Centre (JRC), initiated the sample collection phase of the European Super-Sites initiative, implemented through a HaDEA (European Health and Digital Executive Agency) framework contract. This network of strategically located sites monitors

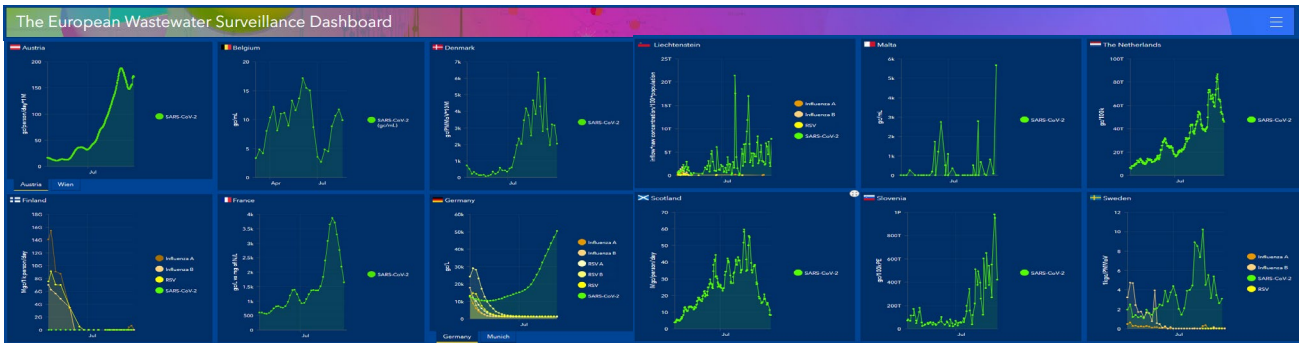
MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

wastewater near transport hubs to detect emerging health threats. As part of the Global Wastewater Sentinel System, the initiative enhances early warning capabilities and supports public health decision-making across the EU and beyond.

SARS-CoV-2 surveillance

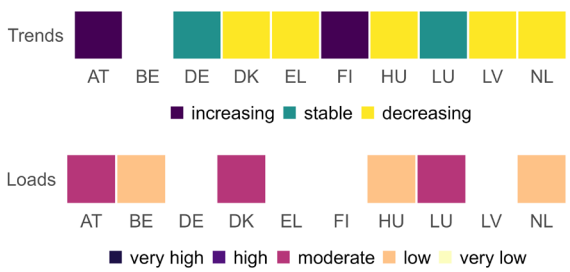
Figure 1. Overview of the European Wastewater Surveillance Dashboard: near real-time wastewater data on measured pathogens for connected countries, providing early warnings and rapid responses to protect public health across Europe.



Source: The European Wastewater Surveillance Dashboard.

A summary of SARS-CoV-2 trends and viral loads is provided (Figure 2) for various EU countries. Figure 2 includes only those countries for which data on viral loads and trends is available.
If information is missing, it indicates that either the national trend or viral load has not been specified, or the data has not been updated to reflect the current month.

Figure 2. Viral loads and trends of SARS-CoV-2 across EU countries.



Source: *EC-JRC*

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

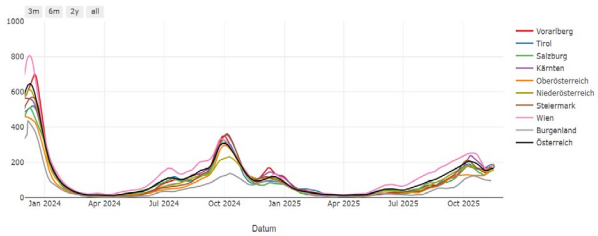
AUSTRIA

Last data update: 17 November 2025

Austria's monitoring of SARS-CoV-2 involves 48 strategically chosen sewage treatment plants, encompassing over 58% of the country's population.

SARS-CoV-2: In week 47, a slight increase in viral loads has been observed, keeping the trend indicator slightly above 1. In week 47, the variant XFG was the most prevalent, including its sub-lineages.

Figure 3. Person-weighted trends in the federal states (and Austria as a whole).



Source*: Abwassermonitoring*.

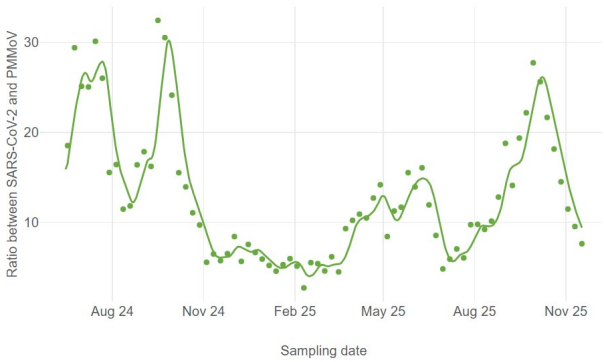
BELGIUM

Last data update: 17 November 2025

The surveillance program encompasses 38% of the Belgian population, involving the collection of wastewater samples once a week from influents of 30 wastewater treatment plants distributed across Belgium, mainly in areas of high population density.

SARS-CoV-2: In week 48 (November 24th) the national level of SARS-CoV-2 was classified as low.

Figure 4. SARS-CoV-2 viral to faecal concentration at a national level.



Source: *Wastewater surveillance Respiratory viruses Sciensano Belgium*.

CZECH REPUBLIC

Last data update: 19 August 2025

Systematic monitoring of wastewater for the presence of SARS-CoV-2 has been conducted since February 2023 at five wastewater treatment plants connected to the largest cities in the Czech Republic and Václav Havel Airport in Ruzyna. The monitored collection points cover approximately 21% of the Czech Republic's population.

SARS-CoV-2: The dashboard, updated to week 40, shows SARS-CoV-2 loads in each treatment plant separately, while at the national level, the presence of the viral load is given as a percentage of positive detections. Currently, this information is only available in Czech.

MONTHLY BULLETIN

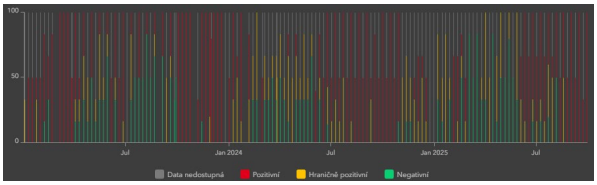
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

CZECH REPUBLIC

Last data update: 19 August 2025

Figure 5. Presence of SARS-CoV-2 viral load in monitored wastewater.



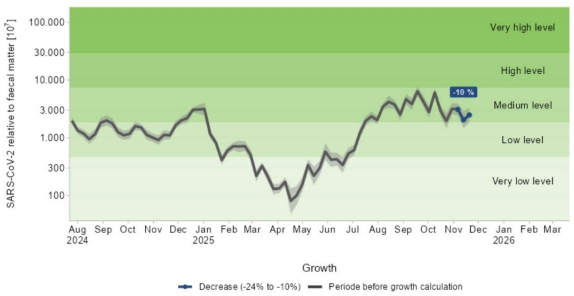
Source: *Monitoring SARS-CoV-2 concentrations in wastewater*

DENMARK

Last data update: 26 November 2025

Denmark's national surveillance of SARS-CoV-2 in wastewater covers 49% of the population. Twice a week, samples are collected from 29 wastewater treatment plants. The results from the wastewater are compiled for the entire country, as well as for the five regions and each specific sampling location.
SARS-CoV-2: In week 47, the concentration of SARS-CoV-2 in wastewater across the country was at a medium level. The national growth rate, reflecting the average weekly growth over the past three weeks, indicates a decrease in the SARS-CoV-2 wastewater concentration (-10%).

Figure 6. Concentration of SARS-CoV-2 in the wastewater, aggregate for all the sampling sites in Denmark.



Source: *National wastewater surveillance of SARS-CoV-2*.

FINLAND

Last data update: 21 November 2025

Various wastewater treatment plants across different regions of Finland actively contribute to the monitoring efforts. Presently, about 44% of the wastewater generated by the Finnish population is included in the wastewater monitoring initiative.
SARS-CoV-2: The results of monitoring coronavirus in wastewater are available at both local and national levels, updated until week 47. The method of determination changed at the end of 2024. A statistical model estimates the trends, which reveals a stable trend in recent weeks.

MONTHLY BULLETIN

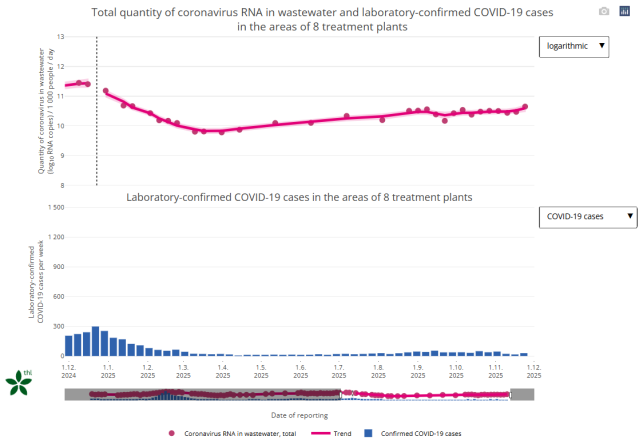
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

FINLAND

Last data update: 21 November 2025

Figure 7. Total quantity of coronavirus in wastewater and laboratory-confirmed COVID-19 cases, in the areas of 8 treatment plants.



Source: *Coronavirus wastewater monitoring weekly report*.

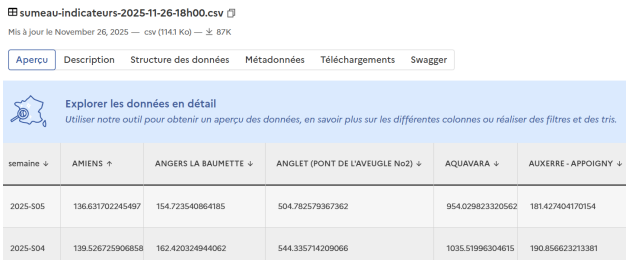
FRANCE

Last data update: 26 November 2025

The national wastewater monitoring system, SUM'Eau, was launched by the General Directorate of Health and the Directorate for Water and Biodiversity, with expert support from health agencies. The data allows for estimating the SARS-CoV-2 circulation trend in 54 treatment plants across 12 administrative regions.

SARS-CoV-2: The data has been updated to week 37 and is accessible in the form of CSV files. The SARS-CoV-2 monitoring indicators in wastewater are computed as the ratio between the viral concentration of SARS-CoV-2 and the concentration of ammoniacal nitrogen. The overall national indicator is also available.

Figure 8. Structure of the CSV file available containing Sars-Cov-2 wastewater surveillance of data.



semaine	AMIENS	ANGERS LA BAUMETTE	ANGLET (PONT DE L'AVEUGLE No2)	AQUAVARA	AUXERRE - APPOIGNY
2025-S05	136.631702245497	154.723540864185	504.782579367362	954.029823320562	181.427404170154
2025-S04	139.526725906858	162.420324944062	544.335714209066	1035.51996304615	190.856623213381

Source: *Data Explorer data.gouv.fr*.

GERMANY

Last data update: 27 November 2025

The viral load of SARS-CoV-2 in wastewater is currently assessed at 169 wastewater treatment plants, providing a coverage for approximately 32% of the population. The number of reporting locations fluctuates over time.

SARS-CoV-2: During the past 4 weeks (reporting week 47), average SARS-CoV-2 viral load measured at wastewater treatment plants has remained stable.

Figure 9. Mean viral load in wastewater.



Source: *Infection Radar*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

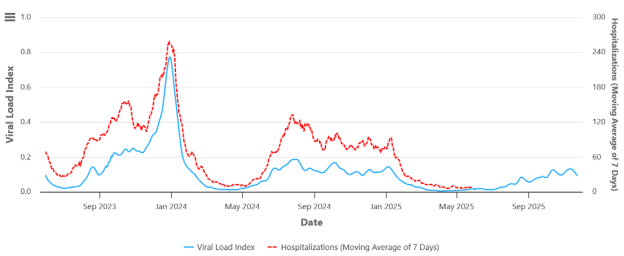
GREECE

Last data update: 30 November 2025

The dashboard presents data from the National Wastewater Epidemiology Network (NWEN), which monitors SARS-CoV-2 and other respiratory viruses in urban wastewater across 10 regions in Greece, covering 70% of the population. It visualises weekly viral load trends at both city and national levels using maps and charts, based on frequent sampling and analysis by coordinated laboratories. The dashboard also includes data on pharmaceuticals and chemicals found in wastewater, offering a comprehensive tool for public health surveillance and early warning.

SARS-CoV-2: During week 48, the national viral load was medium, showing a -18.22% change compared to the previous week, which indicates a declining trend. Among the 8 monitored Wastewater Treatment Plants, 0 displayed an increasing trend, 7 showed a decreasing trend, and 1 remained stable.

Figure 10. National index variation in viral load levels.



Source: *Wastewater surveillance dashboard*.

HUNGARY

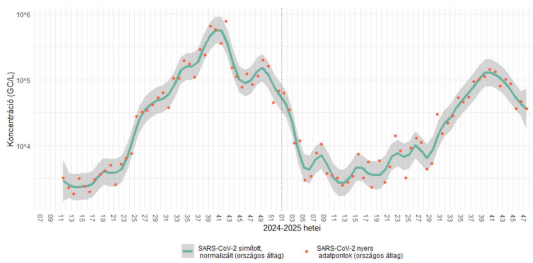
Last data update: 27 November 2025

The data for wastewater monitoring in Hungary are obtained from three wastewater plants serving the entire area of Budapest and a portion of the agglomeration. Additionally, 19 wastewaters from five other agglomeration settlements are examined as a combined sample. Collectively, these analyses encompass over 40% of the domestic population.

The weekly report is currently only available in Hungarian.

SARS-CoV-2: During week 47, the national average concentration of SARS-CoV-2 in wastewater was generally stable or decreasing. From week 37, wastewater analysis detected the XFG variant as predominant, alongside occasional findings of Nimbus, LP.8.1, JN.1, and other X-recombinants.

Figure 11. National, population-weighted average of SARS-CoV-2 copy number measured in wastewater. The red dots show the raw data, the grey band indicates the 95% confidence interval.



Source: *National Center for Public Health and Pharmacy*

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

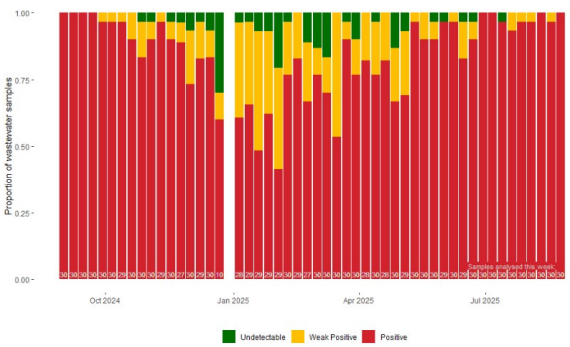
SARS-CoV-2 surveillance

IRELAND

Last data update: 18 October 2025

Since January 2023, Ireland's National Wastewater Surveillance Programme has been conducting weekly sampling in 30 wastewater catchment areas. The 30 catchment areas, covering 70% of the population connected to public wastewater treatment facilities, are distributed across all regions of Ireland.
SARS-CoV-2: In week 35, samples were collected and analysed from 29 out of 30 catchment areas. Of the 29 catchment areas for which a change from the previous week could be calculated, the viral load remained stable (within +/- 10%) in 19 areas, while 1 experienced decreases of 10% or more, and 9 saw increases of at least 10%. SARS-CoV-2 RNA was detected in all the catchment areas.

Figure 12. Wastewater SARS-CoV-2 detections in Ireland. Week 35 2025 (24/08/2025 to 30/08/2025).



Source: *Report week 35.*

ITALY

Last data update: 04 April 2025

Wastewater monitoring in Italy is conducted through Flash Surveys, comprising monthly sampling campaigns at various locations across the country over a short duration.
SARS-CoV-2: The monitoring results are usually reported every two months, with an average delay of a few months from sample collection to reporting. The latest update is from April 2025, where a total of 107 wastewater samples were collected from 16 regions and 2 autonomous provinces (A.P.). Mutations typical of the Omicron JN.1 lineage and its sublineages were identified through Sanger sequencing in three regions, while sequencing data were unavailable for the others.

LATVIA

Last data update: 7 November 2025

The national monitoring program in Latvia focuses on identifying the presence of SARS-CoV-2 in wastewater across municipalities.
Twice a week, untreated wastewater samples are gathered from various facilities within municipal centralised wastewater collection systems.
The weekly is currently only available in Latvian.
SARS-CoV-2: In week 45 the overall trend of RNA concentration in Latvia was moderately decreasing. Detailed reports on the prevalent variants are available for each wastewater facility, revealing that Omicron KP.3 sub-lineage was the most common variant in all locations.

MONTHLY BULLETIN

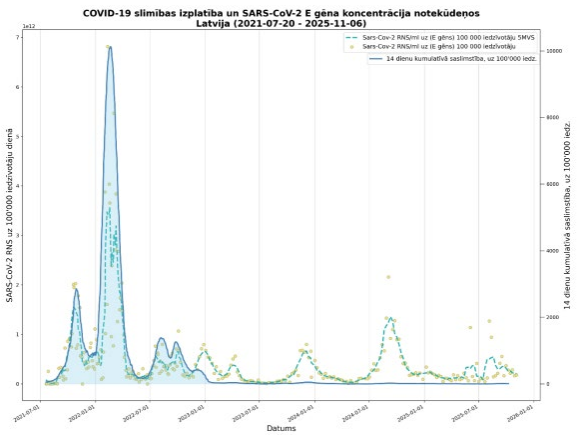
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

LATVIA

Last data update: 7 November 2025

Figure 13. Spread of COVID-19 and detection of SARS-CoV-2 in wastewater.



Source: Wastewater monitoring – Latvia.

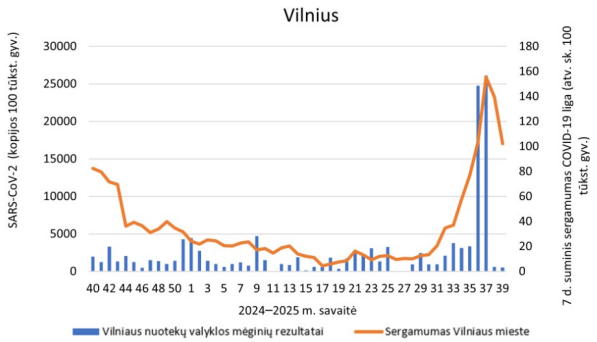
LITHUANIA

Last data update: 28 September 2025

Lithuania has implemented a wastewater surveillance system to detect the SARS-CoV-2 virus and its variants since November 2022. Wastewater samples are collected weekly from three treatment plants.

SARS-CoV-2: Three wastewater treatment facilities are currently under surveillance. The website offers data for each municipality individually, and the latest updated data is from week 39.

Figure 14. Vilnius City Wastewater Treatment Plant (copies/ 100 000 pop.)



Source: Monitoring results 2024-2025 - National public health centre under the Ministry of Health.

LUXEMBOURG

Last data update: 24 November 2025

The program for monitoring the evolution of COVID-19 in wastewater in Luxembourg involves the collection of samples from eight strategically located wastewater treatment plants, covering five cantons and approximately 67% of the country's population. Results are published in fortnightly reports.

SARS-CoV-2: SARS-CoV-2 concentrations are reported at both the national and individual plant levels. In week 48, the national levels of SARS-CoV-2 are moderate.

MONTHLY BULLETIN

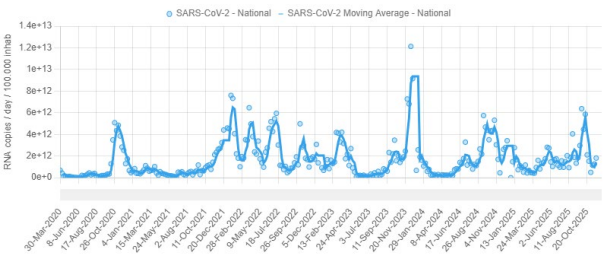
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

LUXEMBOURG

Last data update: 24 November 2025

Figure 15. SARS-CoV-2 national levels (RNA copies/day/100.000 inhab).



Source: SARS-CoV-2 – MicrObs.

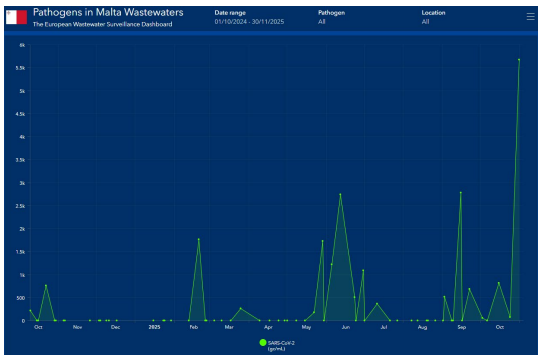
MALTA

Last data update: 30 November 2025

Malta does not currently have a dedicated national dashboard for monitoring SARS-CoV-2 levels in wastewater. However, the Ministry for Health and Active Ageing in Malta regularly submits updates on these measurements to the EU Wastewater Observatory for Public Health. Samples are collected from three wastewater treatment plants.

SARS-CoV-2: SARS-CoV-2 concentrations are reported at both national and individual plant levels. The latest results are updated to week 48.

Figure 16. Malta Dashboard.



Source: Pathogens in Malta wastewaters.

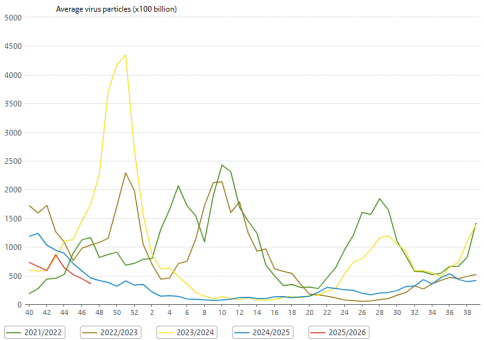
NETHERLANDS

Last data update: 26 November 2025

Wastewater samples are gathered at wastewater treatment plants throughout the Netherlands, ensuring comprehensive coverage.

SARS-CoV-2: During week 48, the national average levels of SARS-CoV-2 in wastewater increased (+45%) compared to the previous week. Multiple JN.1 sub-variants are currently being found, with subvariant XFG being the most common.

Figure 17. The Netherlands. Average virus load (x100 billion) over time per 100,000 inhabitants.



Source: Weekly coronavirus SARS-CoV-2 figures | RIVM.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

POLAND

Last data update: 29 January 2025

The dashboard is supported by a network of voluntary laboratories initiated by the National Institute of Public Health – National Institute of Hygiene (NIZP PZH - PIB). This network includes Provincial Sanitary and Epidemiological Stations, as well as local water supply and sewage companies. The SARS-CoV-2 wastewater data are normalised using PMMoV and reported for four cities, encompassing a total of 15 WWTPs.

SARS-CoV-2: Aggregated city data and specific site data are reported in plots. SARS-CoV-2 mutations detected in wastewater samples are reported for each city. The latest update is from the end of January 2025, and no further updates have been provided since the last monthly bulletins.

SLOVAKIA

Last data update: 3 April 2025

Wastewater surveillance for the presence of SARS-CoV-2 is conducted on a weekly basis in each region of Slovakia, with over 55 wastewater treatment plants being part of the project. Throughout the year, the monitoring intensity is adjusted based on the prevailing epidemiological situation.

SARS-CoV-2: The latest update is from March 2025, and no further updates have been provided since the last monthly bulletin.

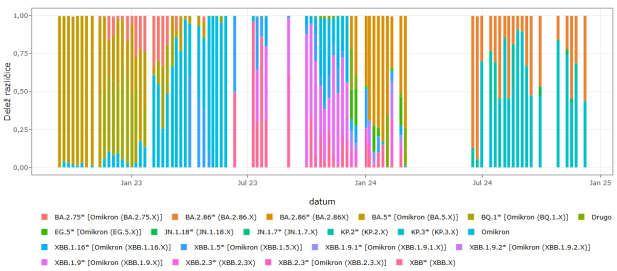
SLOVENIA

Last data update: 11 September 2025

From autumn 2022 and throughout 2024, Slovenia has been consistently monitoring SARS-CoV-2 in wastewater, using samples from 16 waste treatment plants across the nation. This monitoring network reaches about 34% of the Slovenian population and involves weekly sampling.

SARS-CoV-2: The dashboard displays SARS-CoV-2 wastewater monitoring data, including the percentage of circulating virus variants, for each individual wastewater treatment plant. The latest update is from September 2025, and no further updates have been provided since the last monthly bulletin. Results can be accessed via the Slovenian open data portal.

Figure 18. Mutation rates of the SARS-CoV-2 genome.



Source: C19-water.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

SPAIN

Last data update: 22 June 2024

Wastewater surveillance in Spain involves the examination of the presence and evolution of the genetic material of SARS-CoV-2 RNA and the predominant variants in all treatment plants, totaling 54 facilities distributed throughout the country.

SARS-CoV-2: The summary report and dashboard have not been updated since week 25 of 2024.

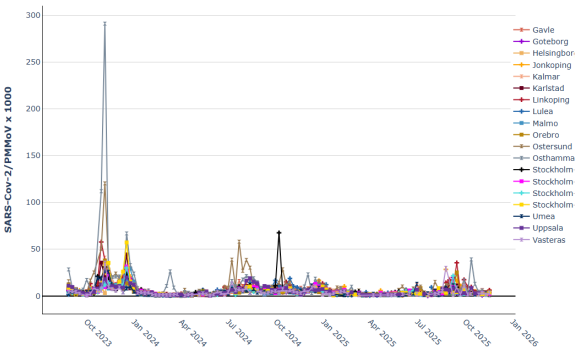
SWEDEN

Last data update: 24 November 2025

The dashboard assesses SARS-CoV-2 levels in wastewater across various Swedish cities. For weekly SARS-CoV-2 monitoring, Sweden's public health agency collaborates with SEEC-SLU (the Swedish University of Agricultural Sciences), providing data covering 43% of the Swedish population with the broadest geographic reach across the country.

SARS-CoV-2: The levels of SARS-CoV-2 in wastewater are visualised using interactive plots. The SEEC-SLU project provides weekly data updated to week 47, which can also be downloaded as a dataset.

Figure 19. PMMoV normalised levels of SARS-CoV-2 virus in multiple cities across Sweden.



Source: SARS-CoV-2 quantification | The Swedish Pathogens Portal.

REGIONAL EU DATA

The EU member states listed below furnish local or regional data regarding the surveillance of SARS-CoV-2 in wastewater for specific areas within their respective countries. The accompanying list includes accessible databases, along with their latest updates.

Austria

Vienna

Weekly wastewater analysis from the Vienna sewage treatment plant is part of the 'CSI Wastewater' project. The graphic shows the weekly occurrence of SARS-CoV-2. Data is updated to week 47, 2025.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

REGIONAL EU DATA

Germany

Berlin

The dashboard presents wastewater monitoring data from 4 treatment plants in Berlin. It shows SARS-CoV-2 data results in gene copies per litre of wastewater sampled from each plant. The analysis also includes circulating virus variants in Berlin, presented as a percentage of the total load. In week 48, the trends indicate a decrease in virus levels in three plants and an increase in one plant.

Bavaria

The dashboard shows wastewater monitoring data from the German state of Bavaria.
It includes the SARS-CoV-2 trend, viral loads and variants in wastewater from thirty municipalities within Bavaria. In week 48, measurements were conducted at 21 locations and the virus activity for SARS-CoV-2 is moderate.

Poland

The website provides surveillance data on the SARS-CoV-2 virus detected in sewage within the Warsaw agglomeration. The findings are presented through graphs that illustrate the normalised number of viral copies at four specific locations within the sewage network. The data is updated weekly, with results reflecting data up to week 48. An increase in virus levels was recorded at all sampling points since August. A particular increase in virus content was observed in mid-October at one location, marking the year's highest level, followed by a sharp decline at the end of the month. November 2025 was marked by rapid changes in virus levels, particularly at one location.

Spain

Madrid

The website presents visualisations of SARS-CoV-2 data in wastewater for the Community of Madrid. A map displays the progression of SARS-CoV-2 levels in wastewater across various municipalities and districts, but it is not updated monthly; the last update was in April 2025. Reports are available in Spanish.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

EU NEIGHBOURING COUNTRIES

This section provides wastewater monitoring details from neighbouring countries of the EU participating as observers in the EU National Contact Points (NCP) Mechanism. Links to national or local dashboards/websites and information about the latest updates are included.

Türkiye

Last data update: 13 July 2025

This website provides information on wastewater monitoring data in Türkiye, explicitly reporting SARS-CoV-2 viral load results. PDF reports are published for various regions (23 pilot cities), while Istanbul has its dedicated report. Wastewater samples are taken weekly from the city of Istanbul and bi-weekly from the other 23 cities, providing a coverage for approximately 41% of the population.

SARS-CoV-2: The latest update is from July 2025, and no further updates have been provided since the last monthly bulletin.

Figure 20. SARS-CoV-2 viral load distribution map, Istanbul city.



Source: COVID-19 Türkiye.

OTHERS

This section contains details on wastewater monitoring from non-EU countries. It includes links to national or local dashboards/websites and the latest updates, organised by continent.

EUROPE

Scotland

Last data update: 23 November 2025

The program has been operating in Scotland since June 2020. Each week, Scottish Water collects approximately 150-200 sewage samples from 115 wastewater treatment facilities across the country. These samples are then sent to NHS Lothian for testing (carried out by the Scottish Environmental Protection Agency until end July 2024).

SARS-CoV-2: Data is available at the national level, as well as for NHS Health Boards and local authority areas. The dashboard is updated to week 48.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

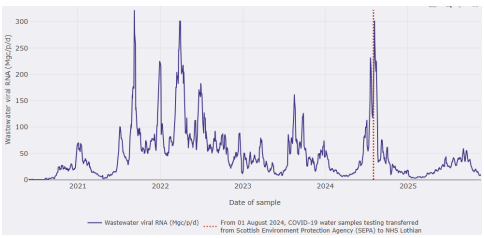
OTHERS

EUROPE

Scotland

Last data update: 23 November 2025

Figure 21. Seven-day average national trend in COVID-19 wastewater.



Source: COVID-19 & Respiratory Surveillance.

Switzerland

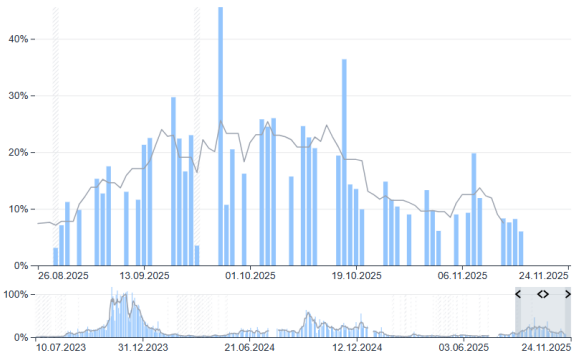
Last data update: 24 November 2025

The information published in the dashboard is based on data from Switzerland's national wastewater monitoring program, which has been tracking the viral load of SARS-CoV-2 since January 2022. As of January 2025, wastewater monitoring has been streamlined to include only 10 WWTPs in Switzerland, covering almost a quarter of the Swiss resident population, and one WWTP in the Principality of Liechtenstein. Additionally, variant monitoring is carried out at six WWTPs from December 2024.

SARS-CoV-2: Each region reports both the relative and absolute viral loads, as well as trends and variants.

During week 48, it was observed that viral loads remained stable in all 11 regions. The most frequently identified SARS-CoV-2 variant (latest update on 05.11.2025) was XFG in all the regions.

Figure 22. Relative viral load of SARS-CoV-2 at a selected location (Lugano).



Source: IDD - Infectious Diseases Dashboard.

AFRICA

South Africa

Last data update: 26 November 2025

The National Institute for Communicable Diseases (NICD) Centre for Vaccines and Immunology provides updated information on SARS-CoV-2 levels in wastewater across 14 districts in South Africa.

SARS-CoV-2: The dashboard presents SARS-CoV-2 levels both nationally and by district. Data is updated up to week 48. As of the beginning of October 2025, the most prevalent sub variant was XFG.X at 85%.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

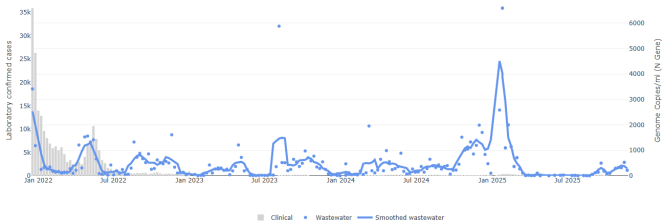
OTHERS

AFRICA

South Africa

Last data update: 26 November 2025

Figure 23. National SARS-CoV-2 wastewater levels.



Source: https://wastewater.nicd.ac.za/.

ASIA

India

Last data update: 19 October 2025

The dashboard displays the concentration of SARS-CoV-2 RNA copies per litre of wastewater in five major cities of Gujarat, India. Samples are collected from 40 wastewater treatment plants. The Lineage Prevalence section is also available.

SARS-CoV-2: The latest update on SARS-CoV-2 concentration in wastewater is from October 2025 (week 42).However, information on variants has not been updated since July 2024.

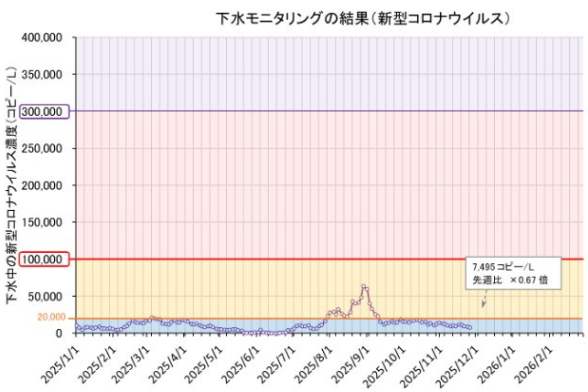
Japan

Last data update: 28 November 2025

The information presented on this website reveals the outcomes of regular assessments of COVID concentration in the wastewater of Komatsu City.

SARS-CoV-2: During week 48, SARS-CoV-2 levels in wastewater were low, showing a stable trend (1.07 times) compared to the previous week.

Figure 24. Virus concentration in Komatsu City expressed in copies/liter.



Source: Sewage Monitoring/Komatsu City Website.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

OTHERS

OCEANIA

Australia

Western Australia

Last data update: 21 November 2025

The dashboard presents an analysis of SARS-CoV-2 levels in the Perth Metropolitan area of Western Australia, starting from July 2022. As of August 2024, wastewater samples are collected weekly from three metropolitan wastewater treatment plants that serve approximately 79% of the city's population.

The concentrations of the virus genome in wastewater are shown as a 14-day moving average and have been updated to week 47. Omicron sub-lineage NB.1.8.X predominated across all three metropolitan wastewater sites (47.2%), followed by BA.3.2.X (20%).

New South Wales Australia

Last data update: 22 November 2025

The New South Wales Sewage Surveillance Program collects samples from 4 sites in Greater Sydney and Newcastle every week to monitor COVID-19 concentrations. The results are published in weekly reports. During week 47, gene concentrations per 1,000 people have been low in all the catchment areas.

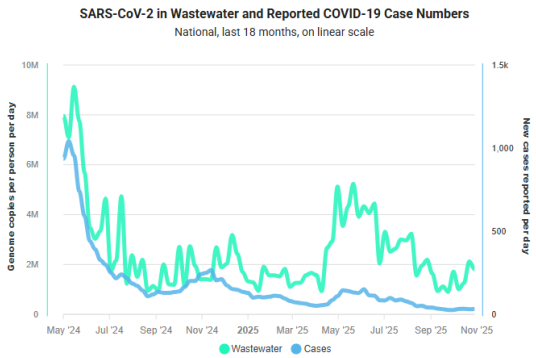
New Zealand

Last data update: 23 November 2025

The dashboard was launched in 2020 to help the public monitor potential COVID-19 risks in their local areas using easy-to-understand data visualisations. Starting in August 2024, data are updated every two weeks, and information on variants is updated at least once a month. Sites and frequency may change over time. Beginning in July 2024, variant analysis is conducted across all wastewater sites.

SARS-CoV-2: SARS-CoV-2 concentrations in wastewater are being monitored at both national and regional levels. In week 47, 11 sites covering 45% of the population tested positive for SARS-CoV-2. In week 47, the dominant strain were the NB.1.8.1 (42%), MC.10.2.1 (9.6%) and XFG (23%). Trends are visually represented on a map.

Figure 25. Average SARS-CoV-2 genome copies detected per person per day in wastewater for New Zealand, along with the reported case numbers.



Source: *ESR Wastewater Surveillance*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

OTHERS

NORTH AMERICA

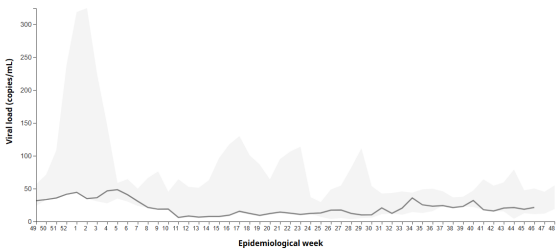
Canada

Last data update: 28 November 2025

The dashboard provides information on the levels of COVID-19 and its variants detected in the wastewater from various communities across Canada. Currently, 44 sites submit sewage samples for processing, representing 31.2% of the population. Additionally, the dashboard includes links to 23 Canadian wastewater COVID-19 surveillance dashboards and websites, which are managed by provincial, territorial, and academic partners across the country.

SARS-CoV-2: In week 48, the COVID-19 activity level in Canada was classified as moderate. 44 sites submit wastewater samples for processing. Of those sites, 12 showed a high activity level, 16 showed a moderate activity level, 10 showed a low activity level, 2 had no detection and 4 were not available.. November 16th shows XFG as the dominant variant identified.

Figure 26. Trend graph of 7-day rolling average of COVID-19 viral load in wastewater.



Source: Wastewater monitoring dashboard – Canada.ca

United States

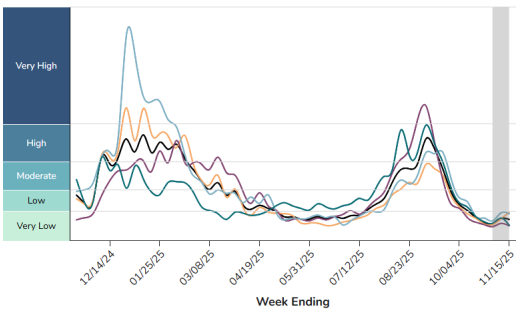
Last data update: 15 November 2025

The National Wastewater Surveillance System (NWSS) collects and analyses wastewater samples on a weekly basis. This monitoring is conducted in all 50 states, 7 territories, and some tribal communities. Over the past two months, 1,341 sampling sites have reported data to the NWSS, representing approximately 44% of the U.S. population.

SARS-CoV-2: During week 44, the national viral activity level for COVID-19 in wastewater was very low.

The predominant variant during this week was XFG, which accounted for 90% of the total abundance.

Figure 27. COVID-19 Wastewater Viral Activity Level Over 1 Year Time.



Source: COVID-19 Wastewater Data – National Trends | NWSS | CDC.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Mpox surveillance

This section provides a comprehensive overview of recent mpox outbreaks with a focus on wastewater surveillance and related news from around the world.

On September 5, 2025, the World Health Organization (WHO) officially declared the end of the global Mpox emergency, following sustained reductions in case numbers and mortality rates, particularly in the Democratic Republic of Congo, Burundi, Sierra Leone, and Uganda. Despite this milestone, Mpox remains a significant public health concern, especially in Africa, which continues to account for over 90% of reported cases. According to the WHO's latest situation report (dated September 19), a total of 3,780 confirmed cases and 15 deaths were recorded globally in August. Both Clade I and Clade IIb strains remain in circulation.

Mpox transmission in Europe remains low but persistent. Most cases continue to involve Clade IIb. In recent months, several EU/EEA countries—including Germany, France, Belgium, Sweden, and Ireland—have reported isolated Clade I cases, all linked to travel from endemic regions in Africa. The UK confirmed 16 Clade I cases and over 580 Clade IIb cases since 2023, with no evidence of sustained community transmission of Clade I. Berlin experienced a rise in Clade IIb cases earlier in the year, prompting renewed public health outreach. Overall, although Europe has seen some cases, the situation remains under control due to ongoing surveillance and preventive measures.

Mpox wastewater surveillance

This section provides information on mpox wastewater surveillance derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities.

EU Countries

Spain

Within the framework of the Spanish Wastewater Surveillance Programme, in December 2024 and January 2025, 10 different sampling points were assessed and 100% of them tested negative for mpox by RT-qPCR assays. The team is still working on returning to take weekly samples on 66 sampling points, due to some funding matters, and assess again routine SARS-CoV-2 samples, adding mpox, enterovirus and poliovirus to the monitored agents of interest list.

Non-EU Countries

Canada

The wastewater dashboard summarises the consistent detection of mpox virus signals in wastewater samples from selected cities across Canada. Signal interpretation is based on data from only four weeks of testing. Recent data from week 48 indicate no detection across 5 cities. In contrast, there was consistent detection, defined as detection of mpox for more than 2 weeks in the past 4 weeks, in one city, Toronto.

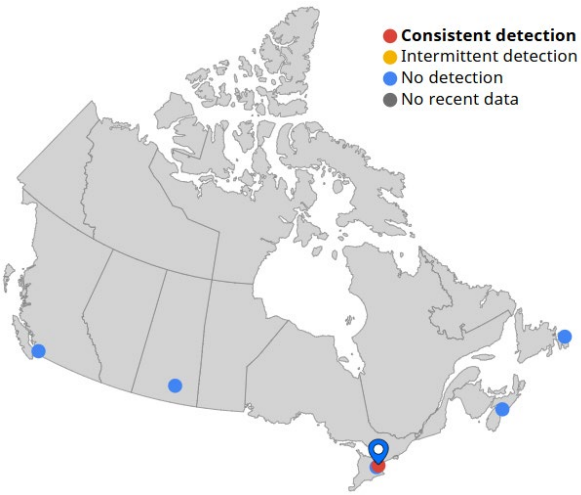
MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Mpox surveillance

Canada

Figure 28. Recent mpox detection in wastewater in cities across Canada.



Source: Mpox activity - Wastewater monitoring dashboard — Canada.ca.

United States

The CDC's National Wastewater Surveillance System (NWSS) provides a public health framework to monitor infectious diseases through wastewater across the country. Wastewater testing for the mpox virus differs by location. Some locations test for specific clades, focusing on either clade II or clade I, while others use methods that detect both clades without distinguishing between them. Communities are voluntarily monitoring the presence of mpox in wastewater on a weekly basis. During the four-week period from October 19, 2025 - November 15, 2025, 6 out of 550 wastewater sites (1%) that tested for mpox reported the presence of the virus. Specifically, 3 sites indicated the detection of clade II, and 1 site indicated the detection of clade I. The remaining sites either reported no mpox detection in wastewater (510 sites) or had no recent data available (40 sites). The remaining sites either reported no mpox detection in wastewater (486 sites) or had no recent data available (44 sites).

Figure 29. Mpox virus detection in wastewater in the past 4 weeks, United States.



Source: U.S. Mpox Wastewater Data | National Wastewater Surveillance System | CDC.

Mpox news

7.11.2025

MPOX outbreak in Africa: the urgent need for local manufacturing of the vaccine and decolonized health systems

14.11.2025

Monkeypox in the United States and Around the World: Current Situation | Monkeypox | CDC

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Mpox news

15.11.2025

Mpox in Kenya - The Lancet

22.11.2025

Mpox due to monkeypox virus clade I infection detected outside of Africa: a case study

27.11.2025

A Collaborative Healthcare Response to The Mpox Epidemic: Lessons from A Nurse–Assistant–Physician Triage and Vaccination Model in France

Avian Flu (H5N1) surveillance

In addition to monitoring avian flu trends through wastewater surveillance data from national dashboards and National Health Authorities' websites, this section provides a comprehensive overview of recent Avian flu news.

Avian Flu in Wastewater: Since the beginning of 2025, avian influenza A(H5N1) has been increasingly detected in wastewater in various regions, especially in the United States. In Europe and worldwide, there has been a continued rise in outbreaks among wild birds and mammals, with new cases emerging even in remote areas, such as the sub-Antarctic islands.

Global Situation: Avian influenza A(H5N1) remains a significant global concern, with widespread outbreaks continuing among wild birds and poultry across multiple continents. According to the WHO, the global public health risk from H5N1 remains low, though occupational exposure risk is considered low to moderate depending on hygiene and biosecurity measures. Human infections are still primarily linked to direct contact with infected poultry or contaminated environments. There is no evidence of sustained human-to-human transmission.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Avian Flu News

13.11.2025

Highly Pathogenic Avian Influenza Viruses (HPAIV) Associated with Major Southern Elephant Seal Decline at South Georgia | Communications Biology

24.11.2025

Avian influenza virus A(H5N1) genotype D1.1 is better adapted to human nasal and airway organoids than genotype B3.13 | The Journal of Infectious Diseases | Oxford Academic

26.11.2025

Characterization of H5N1 high pathogenicity avian influenza virus belonging to clade 2.3.4.4b isolated from Ezo red fox in Japan in a mouse model – PubMed

29.11.2025

Influenza mRNA vaccine reduces pathogenicity and transmission of A(H5N1) virus in a ferret model | npj Vaccines

Avian flu wastewater surveillance

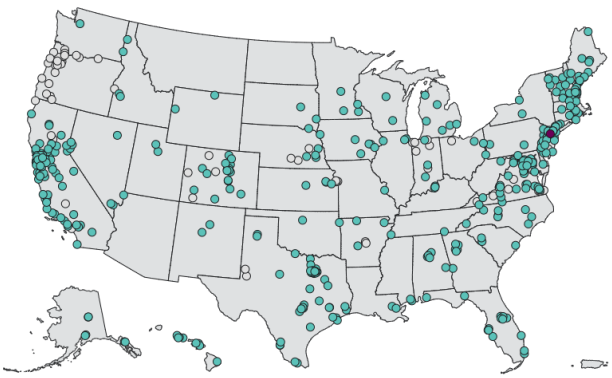
This section provides more recent information, gathered from dashboards and websites owned and operated by National Health Authorities, and listed on the EU Wastewater Observatory for Public Health.

NON-EU countries

United States. The CDC receives weekly data on the avian influenza A(H5) virus from a contract partner, an academic group, and several state health departments. Wastewater sampling and testing are conducted one or more times each week.

During week 45, avian influenza A(H5) results were reported from 432 sites. Of these, 1 site (0.3%) reported H5 detections, while 382 (99.7%) sites reported no detection.

Figure 30. Site-level data for avian influenza A(H5) viral RNA detections in wastewater.



Source: *Wastewater Data for Avian Influenza A(H5) | National Wastewater Surveillance System | CDC*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Poliovirus surveillance

This section provides a summary of latest detected news on poliovirus to keep informed about the most current developments.

Europe: Between September and December 2024, vaccine-derived poliovirus type 2 (cVDPV2) was detected in wastewater samples across multiple European countries, including Germany, Poland, Spain, Finland, and the United Kingdom. These detections are part of ongoing environmental surveillance efforts and no human polio cases have been reported in association with them. These detections primarily involve circulating vaccine-derived poliovirus type 2 (cVDPV2) genetically linked to a lineage first identified in Nigeria in 2020.

As of October 2025, no human cases of polio or poliovirus infections have been reported in association with these environmental detections in the EU/EEA, and the region remains free from the disease. However, the presence of poliovirus in wastewater highlights the need for continued vigilance and preventive measures, especially in under-vaccinated communities.

The European Centre for Disease Prevention and Control (ECDC) emphasises the importance of maintaining robust surveillance systems, including acute flaccid paralysis (AFP) monitoring and environmental surveillance, to rapidly detect and respond to any circulation of poliovirus.

Global update: Polio remains a global health concern, with wild poliovirus type 1 (WPV1) still endemic in Pakistan and Afghanistan, where 13 cases have been reported so far in 2025. Circulating vaccine-derived poliovirus type 2 (cVDPV2) continues to spread across 35 countries, with detections increasingly found in wastewater samples in regions including Europe, Africa, and Asia.

Poliovirus News

15.11.2025

Polio Bulletin (Year 40, 45-46) - 15 November 2025 - PAHO/WHO | Pan American Health Organization

17.11.2025

RKI - Infectious diseases in Germany - Wild poliovirus type 1 detected in wastewater sample in Hamburg

21.11.2025

Statement of the Forty-third meeting of the Polio IHR Emergency Committee

Poliovirus Wastewater Surveillance

This section provides more recent information, gathered from dashboards and websites owned and operated by National Health Authorities, and listed on the EU Wastewater Observatory for Public Health.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Poliovirus Wastewater Surveillance

EU countries

Belgium. Belgium has started monitoring poliovirus in January 2024. The surveillance program involves collecting wastewater samples every four weeks from the influent of 6 wastewater treatment plants. They do not plan to publish poliovirus data in a report.

Finland. Finland has been monitoring poliovirus in wastewater since the 1960s. Samples are collected biweekly in the Helsinki capital region and monthly in four other cities, covering approximately 30% of the Finnish population. At the moment, no data is being reported.

Ireland. Since February 2023, samples from the largest wastewater catchment area in Ireland. have been tested every two weeks for poliovirus 2 (PV2). Currently, the focus is on PV2, based upon recent detections elsewhere of cVDPV2. Additional catchment areas and poliovirus types may be added over the coming months.

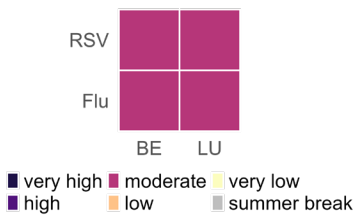
Slovenia. In 2024, Slovenia began monitoring for poliovirus in wastewater to detect any potential reintroduction of the virus into the country. This monitoring is conducted every three months at 16 WWTPs from all statistical regions, covering 34.2% of the population. The latest monitoring results, from September 2025, showed that all samples were negative for the virus.

Influenza, RSV and other viruses' surveillance

EU countries

This section provides information on RSV, influenza and other viruses wastewater surveillance derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities. Figure 30 illustrates the comparative viral loads of Influenza and RSV across various EU countries and includes only those countries for which indications on viral loads are provided.

Figure 31. Viral loads for Influenza and RSV across EU countries.



Source: EC-JRC.

Influenza

EU countries

Austria (regional data). Since October 2022, the influenza virus has been monitored weekly in the city of Vienna. The virus load is depicted in the graphics updated to week 47 and the data, including previous seasons, is available for download as an open governance dataset.

MONTHLY BULLETIN

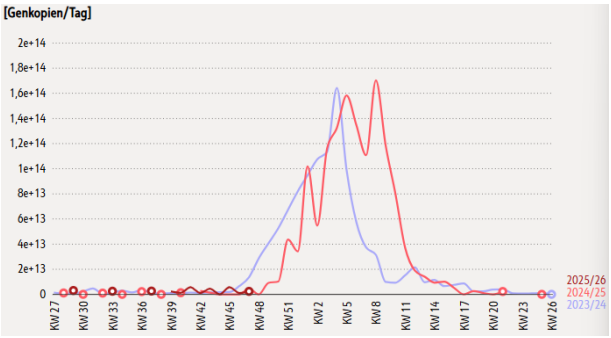
EU Wastewater Observatory for Public Health

Influenza

EU countries

Austria

Figure 32. Influenza virus load in Vienna.

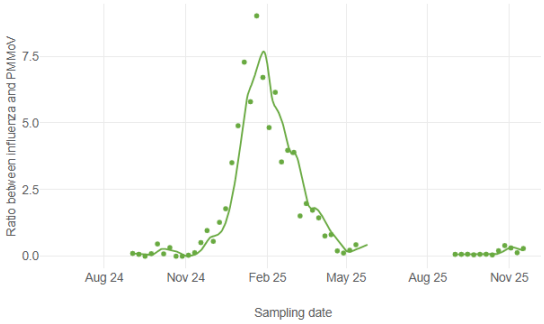


Source: Wastewater Monitoring - Weekly Data.

Belgium

Belgium has been monitoring the presence of influenza A and B since the beginning of this year. The surveillance program includes collecting wastewater samples once a week from the influent of 30 wastewater treatment plants. During week 45, the Influenza concentration was at a moderate level.

Figure 33. Ration between Influenza and PMMoV.

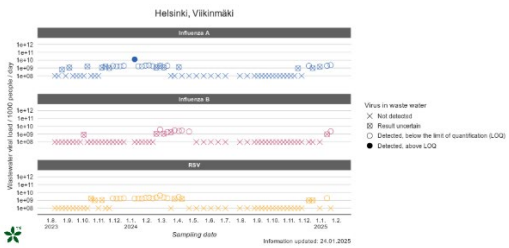


Source: Wastewater Sciensano.

Finland

The results for influenza A and influenza B, starting from July 31, 2023, are available at the municipal level for 11 locations. Quantitative results for 10 of these locations have been updated up to week 45 and can be downloaded in CSV format.

Figure 34. Influenza A and B viral loads in wastewater (top and middle panels), Helsinki.



Source: Influenza A, influenza B, and RSV in wastewater - THL

EU countries

Germany

Influenza A and Influenza B viral loads in wastewater are monitored at several wastewater treatment plants, and the number of reporting locations can fluctuate over time. For calendar week 47, data from 66 WWTPs were available. The average total viral load for Influenza A and B increased in the past 4 weeks. Additionally, monitoring for Influenza A and B in wastewater has been conducted in the German state of Bavaria since the beginning of 2024. Trends for Influenza are reported weekly on the local dashboard. During week 47, the viral load showed a increase in the monitored locations.

MONTHLY BULLETIN

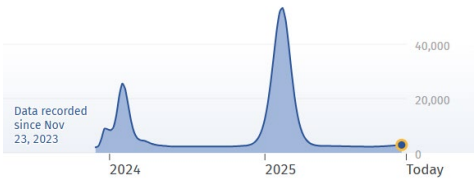
EU Wastewater Observatory for Public Health

Influenza

EU countries

Germany

Figure 35. National influenza viral load in wastewater (gene copies per liter).



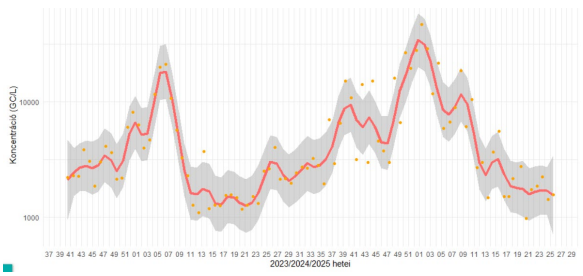
Source: *Infektionsradar*.

Hungary

The Influenza A virus has been detected in wastewater since January 2023. Starting from week 48 of 2024, the results for influenza A were supplemented with the amount of influenza B genetic material detected.

The sampling locations are the same as those used for SARS-CoV-2. The most up-to-date data currently available is from June 26, 2025.

Figure 36. National, population-weighted average of Influenza A copy number measured in wastewater.

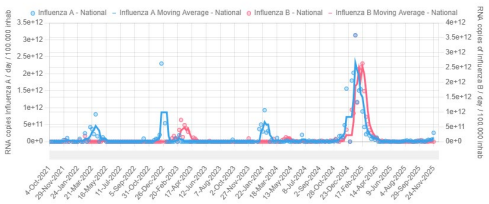


Source: *National Center for Public Health and Pharmacy - 2025.*

Luxembourg

The dashboard monitors the levels of Influenza A and B in wastewater from five cantons in Luxembourg, which represent about 67% of the population and include data from eight treatment plants. Results have been reported since October 2021, with updates every two weeks. As of week 48, the loads of Influenza A and B are moderate.

Figure 37. Influenza A/B national levels (RNA copies/day/100.00inhab).

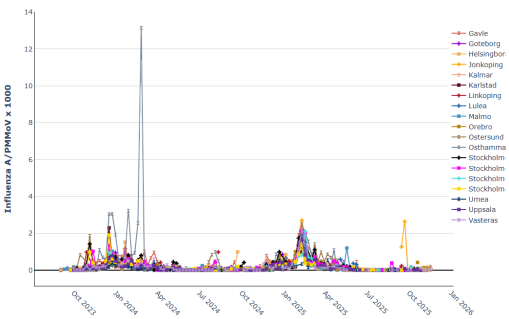


Source: *Influenza – MicrObs*.

Sweden

SLU-SEEC (the Swedish University of Agricultural Sciences) collects and analyses samples for influenza A and B viruses from 24 sites across Sweden, covering 43% of the population. The results are available up to week 45.

Figure 38. PMMoV normalised Influenza A content.



Source: *Amount of influenza virus in wastewater (SLU) | The Swedish Pathogens Portal*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

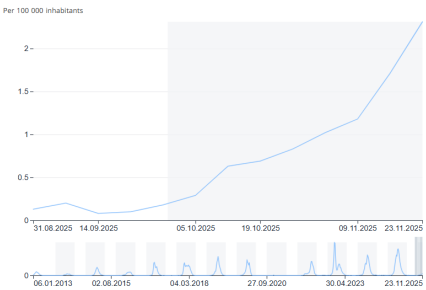
Influenza

Non-EU countries

Switzerland

Influenza A and B have been monitored through the national wastewater monitoring system since July 2023. Starting in January 2025, wastewater monitoring will be conducted at 10 participating WWTPs in Switzerland and one WWTP in the Principality of Liechtenstein. In week 48, the trends for Influenza A and B indicated an increase across all regions.

Figure 39. Development over time of influenza detections.



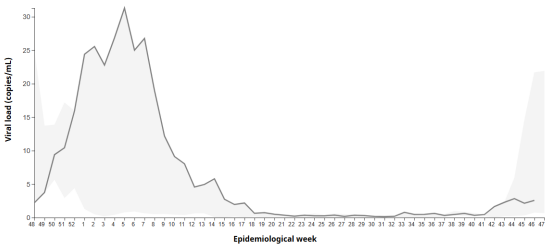
Source: IDD - Infectious Diseases Dashboard.

Canada

The dashboard provides information about the levels of Influenza A and B found in the wastewater from various communities across Canada. In week 48, 19 sites are submitting sewage samples for analysis, which represents 29.2% of the Canadian population. The activity level of Influenza A was classified as moderate. Among the 19 tested sites, 7 showed a high activity level, 10 showed a moderate activity level and 2 showed a low activity level.

For Influenza B, the activity level was classified as low because 1 showed a high activity level, 3 showed a moderate activity level, 1 showed a low activity level and 14 had no detection.

Figure 40. National average of Flu A viral load (copies/ml) in wastewater.



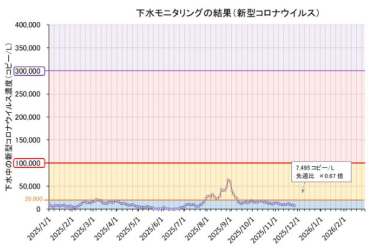
Source: Wastewater monitoring dashboard – Canada.ca.

Japan

In October 2024, Komatsu City began analysing and sharing information about the influenza A virus, providing updates 2-3 times a week. Influenza virus analysis is conducted from October to March.

During week 49, the levels of influenza A were low.

Figure 41. Influenza A virus concentration in Komatsu City expressed in copies/liter.



Source: Influenza A Wastewater Data – Komatsu City

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

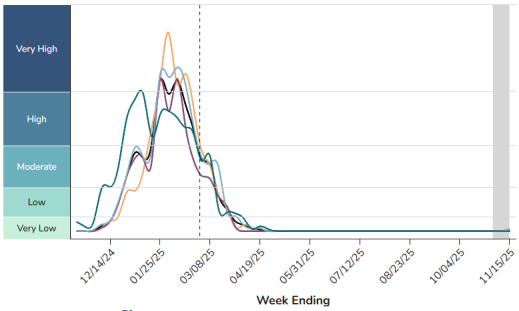
Influenza

United States

The CDC and public health partners are continuing to investigate whether the changing levels of influenza A virus in wastewater are related to seasonal influenza, the avian influenza A (H5) outbreak in animals, or both. Wastewater data cannot determine the specific source of influenza A viruses.

In week 48, the activity level of influenza A in wastewater across the country was very low.

Figure 42. National and Regional trends of wastewater viral activity levels of influenza A.



Source: Influenza A Wastewater Data – National Trends | NWSS | CDC.

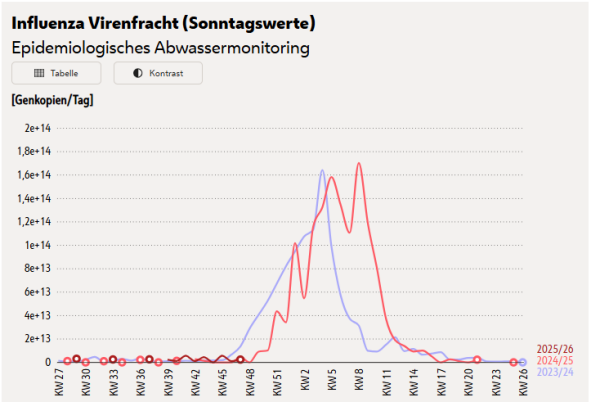
RSV

EU countries

Austria

(regional data). Since October 2022, the RSV has been monitored on a weekly basis in the city of Vienna. The virus load is depicted in the graphics updated to week 47, and the data, including previous seasons, is available for download as an open governance dataset.

Figure 43. Ratio between RSV and PMMoV.

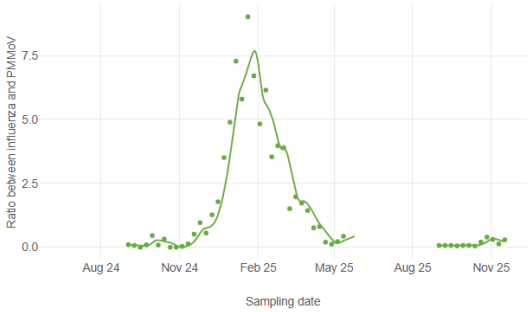


Source: Wastewater Monitoring - Weekly Data.

Belgium

Belgium started monitor RSV since the beginning of this year. The surveillance program involve the collection of wastewater samples once a week from the influent of 30 wastewater treatment plants. During week 45, the RSV concentration was at a moderate level.

Figure 44. Ratio between RSV and PMMoV.



Source: wastewater sciensano

MONTHLY BULLETIN

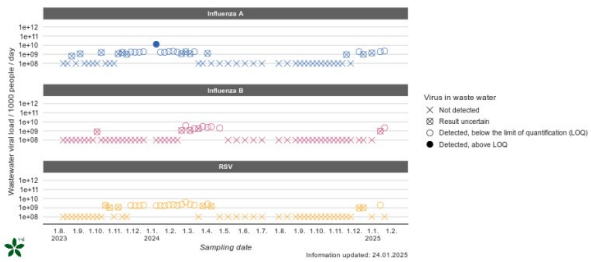
EU Wastewater Observatory for Public Health

RSV

Finland

The results RSV, starting from July 31, 2023, are available at the municipal level for 11 locations. Quantitative results for 8 of these locations have been updated up to week 37 and can be downloaded in CSV format.

Figure 45. RSV viral load in wastewater (bottom panel), Helsinki (Number of residents within the network: 860,000). (information updated: 24.01.2025).



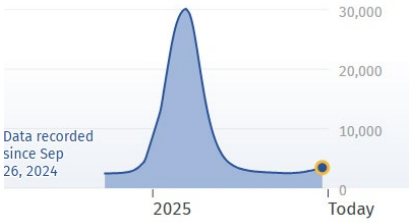
Source: Influenza A, influenza B, and RSV in wastewater – THL.

EU countries

Germany

RSV viral loads in wastewater are monitored at several wastewater treatment plants, and the number of reporting locations can fluctuate over time. During reporting week 45, data from 42 WWTPs were available. The average viral load for RSV increased in the past 4 week. Since the beginning of 2024, wastewater monitoring for RSV has been conducted in Bavaria, Germany. Trends for RSV are reported weekly on the local dashboard. During week 45, the viral load showed a slight increase.

Figure 46. National RSV viral load in the wastewater (no. of gene copies per litre).

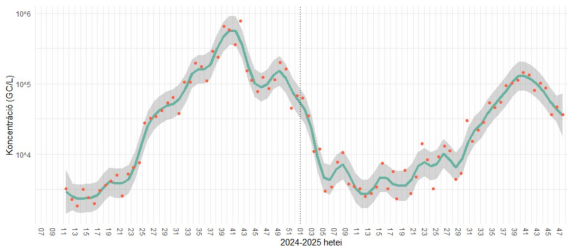


Source: Infection Radar.

Hungary

RSV virus has been detected in wastewater since January 2023. The sampling locations are the same as those for SARS-CoV-2. During week 47, the national average concentration of RSV showed a decrease compared to the previous week.

Figure 47. National, population-weighted average of RSV copy number measured in wastewater.



Source: National Center for Public Health and Pharmacy - 2025. Week 42 wastewater test results.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

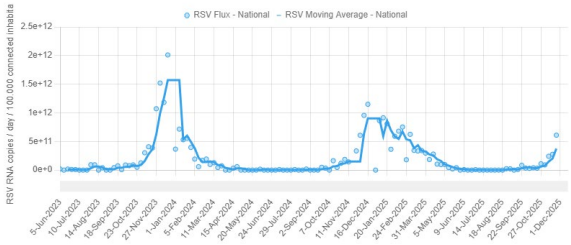
RSV

EU countries

Luxembourg

The dashboard monitors the levels of RSV in wastewater from five cantons in Luxembourg, which represent about 67% of the population and include data from eight treatment plants. Results have been reported since June 2023, with updates every two weeks. As of week 48, RSV levels are moderate.

Figure 48. RSV national levels (RNA copies/day/100.00inhab).

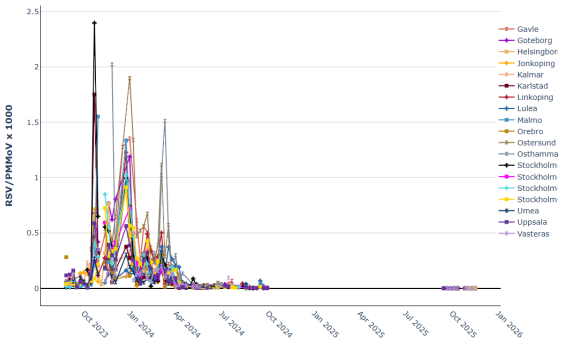


Source: RSV – MicrObs.

Sweden

SLU-SEEC (the Swedish University of Agricultural Sciences) collects and analyses samples for RSV from different areas of Sweden. The results are available up to week 46 for 19 cities.

Figure 49. RSV concentration in copies/liter.



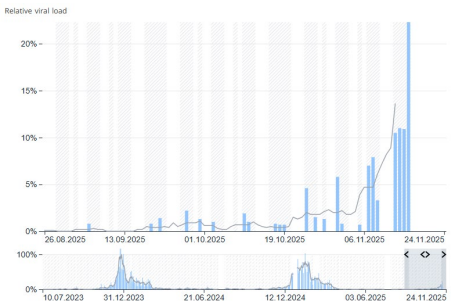
Source: Amount of Respiratory Syncytial Virus (RSV) in wastewater (SEEC-SLU) | The Swedish Pathogens Portal.

Switzerland

RSV has been monitored through the national wastewater monitoring system since July 2023. Starting in January 2025, wastewater monitoring takes place at 10 participating WWTPs in Switzerland and one WWTP in the Principality of Liechtenstein.

As of week 48, the RSV trends have been stagnant across 10 regions and 1 region showed a low level.

Figure 50. RSV relative viral load for a selected region (Lugano).



Source: IDD - Infectious Diseases Dashboard.

Canada

The dashboard presents data on the levels of RSV detected in the wastewater of various communities across Canada. Currently, 19 sites are submitting sewage samples for analysis, representing 29.2% of the Canadian population. In week 42, the RSV activity level was classified as moderate. 1 site submitted wastewater samples for processing, representing 25.5% of the population in Newfoundland and Labrador.

MONTHLY BULLETIN

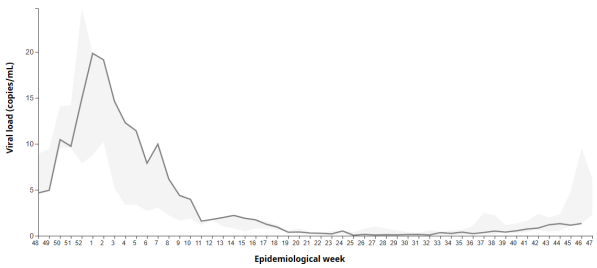
EU Wastewater Observatory for Public Health

RSV

Non-EU countries

Canada

Figure 51. National average of RSV viral load (copies/ml) in wastewater.



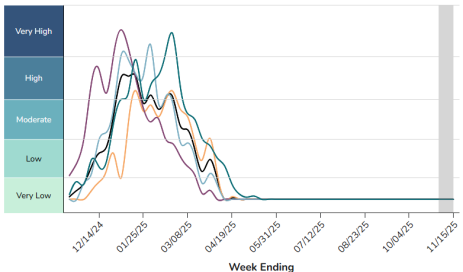
Source: Wastewater monitoring dashboard – Canada.ca.

United States

CDC's National Wastewater Surveillance System (NWSS) provides the public health infrastructure to monitor RSV through wastewater across the country.

During week 45, the level of viral activity for RSV in the wastewater was very low on a national level.

Figure 52. National and Regional trends of wastewater viral activity levels of RSV.



Source: RSV Wastewater Data – National Trends | NWSS | CDC.

Enteric Viruses

EU countries

Luxembourg

Norovirus and Enterovirus have been monitored in Luxembourg from March 2020 to July 2024. Wastewater levels were reported from two cantons, covering approximately 52% of the population through four treatment plants. Latest sampling date: 15.07.2024.

Sweden

Enteric viruses, including enteroviruses, adenoviruses, GG2, astroviruses, sapoviruses, and pepper mottle mild virus, have been monitored in Gothenburg by the University of Gothenburg, However, as of April 2024, the data is no longer being updated but is being kept for historical reference.

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Blue Mirror

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